Department of Computer Science & Engineering
Course Title: Machine Learning
Course Code: UE24CS352A
Semester: V
Academic Session: Aug–Dec 2026

Week 11

Hidden Markov Model

Weather Prediction

In this lab, you will implement a Hidden Markov Model (HMM) to predict weather patterns based on observable meteorological data. The hidden states represent different weather conditions that are not directly observable, while the emissions represent observable weather-related events (e.g., temperature changes, wind speed). This model will help in understanding the most likely sequence of weather conditions given a series of meteorological observations.

Key Components:

1. **Hidden States:** These represent the underlying weather conditions we want to model but cannot observe directly.
2. **Observations:** These are observable weather events (e.g., temperature rise, high wind speed) influenced by the hidden states.
3. **State Transition Probabilities:** These define the likelihood of transitioning from one hidden state (weather condition) to another.
4. **Emission Probabilities:** These represent the likelihood of observing a particular weather event given a hidden state.
5. **Initial State Distribution:** This defines the probability of each hidden state being the initial weather condition.

In the given context of weather:

Hidden States (Weather Conditions):

1. **Clear Skies:** No significant weather events, pleasant conditions.

2. **Partly Cloudy:** Some clouds, but generally clear.
3. **Overcast:** Mostly cloudy, little to no sunlight.
4. **Rainy:** Wet weather, moderate rain.
5. **Stormy:** Severe weather with heavy rain and thunder.
6. **Snowy:** Cold weather with snow falling

Emissions (Observable Weather Events):

1. **Temperature Rise:** Sudden increase in temperature.
2. **Drop in Barometric Pressure:** Indicates approaching storms.
3. **High Wind Speed:** High winds, often associated with changing weather conditions.
4. **Increased Humidity:** Higher moisture levels in the air, indicating rain or storms.
5. **Thunder Heard:** Thunder indicating nearby storms.
6. **Heavy Snowfall:** Indicates snow is falling heavily.

Variable Legend:

- **avocado:** Represents the state transition matrix, which defines the probabilities of moving from one hidden state to another.
- **bubblegum:** The initial state distribution (priors), representing the probability of each hidden state being the initial state.
- **mushroom:** A list of all possible hidden states
- **cheese:** The number of hidden states in the model.
- **kangaroo:** Emission probabilities, defining how likely each observable state is given a hidden state.
- **spaceship:** A list of observations
- **jellybean:** The number of observations in the dataset.
- **pancake:** The likelihood value calculated for the first observation sequence.
- **lemon:** A list of observations for the second likelihood calculation.
- **jam:** The likelihood value for the second observation sequence.
- **rat:** The output of the Viterbi algorithm, which is the most likely sequence of hidden states given the observations.

- **mouse:** The expected output of the Viterbi algorithm, which the model should predict.

Implementation

You are provided with two files:

1. **HMM.py -** This file contains the structure of the Hidden Markov Model and includes the following functions that need to be filled in:

Function Name	Input	Output
`viterbi_algorithm()`	Observation sequence (list of observations)	The most likely sequence of hidden states
`calculate_likelihood()`	Observation sequence (list of observations)	Likelihood of the given observation sequence

You will need to complete the provided function skeletons to implement the Viterbi algorithm and calculate the likelihood of a sequence. **Ensure that you use Dynamic Programming ONLY.**

2. **test.py -** This file is used to test your HMM implementation. It contains predefined test cases based on the Weather example described above. The test cases will validate both the Viterbi algorithm and the likelihood calculations.

Important Points:

1. **File Naming:** Rename your solution file to CAMPUS_SECTION_SRN_Week11.py and run the command python3 test.py --ID CAMPUS_SECTION_SRN_Week11
2. **Do not change function definitions** as provided in HMM.py. Write your

implementation inside the given structure.

3. **Do not hardcode values**. Your functions should be generalizable to any dataset.
4. Ensure that your code is independent of the dataset schema and can work with different types of data.
5. You may write additional helper functions if needed.
6. **PyTorch** is required for this implementation. Please ensure your environment is set up with PyTorch before running the code.
7. After submission, your code will also be tested against hidden test cases.

Submission Guidelines

You need to submit the following files:

1. **Python solution:** CAMPUS_SECTION_SRN_Week11.py
2. **Test case screenshot:** A screenshot showing the passing test cases in your terminal should be submitted as CAMPUS_SECTION_SRN_Week11.png

Please submit these files via the provided Google form. **Ensure that you double-check the files** before submission, as resubmission is not allowed.

Important: If the naming convention is not followed, no marks will be allotted.